Note Title 9/3/2025

COL774

Sep 3, 2025

Bayesian Network

$\forall j: X_j \mid Pa(X_j)$

$\{X_j \perp \text{Non-Desc} \mid Pa(X_j)\}$

$\Downarrow$

independences encoded by a Bayesian network

↳

Naive Bayes Model

Cough $\perp$ Cold | Disease

↳ $\underbrace{X_j \perp X_k \mid Y}$

↳ Naive Bayes

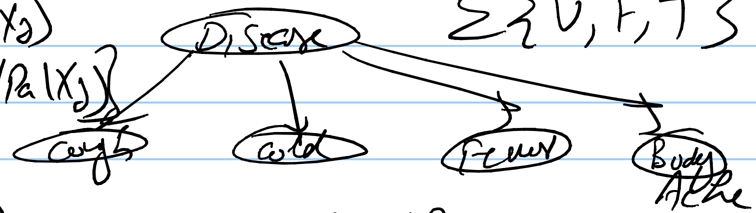
$\forall j \Rightarrow X_j \mid Y$

For Naive Bayes

↳ (1) $Y \sim$ Bernoulli(ϕ) | Multinoulli(Φ)

Graphical Models

$\in \{V, F, T\}$



$\in \{0, 1\}$

$\underbrace{X_1 \,--\, X_n, \, Y}$

↳ Naive Bayes

$X_j \perp X_k \mid Y$

② $\forall j \quad X_j \mid Y \sim \text{Multinoulli}(\theta_{j|y})$

$\theta_{j|y} \Downarrow \in R^{r}$

$\theta = (\Phi, \theta)$ → parameters of conditional

↳ prior over y

dist. $X_j \mid Y$

$X_j \in \{1 \,--\, r\}$

$\Rightarrow \sum_{\ell} \theta_{j\ell|y} = 1$

Estimate the parameters :-

$\underset{\theta}{\arg\max}\, \mathcal{LL}(\theta) =$

$\underset{\theta}{\arg\max} \log \left[\prod_{i=1}^{m} P(x^{(i)}, y^{(i)}; \theta) \right]$ i.i.d over samples

\# parameters $= c + nrc$

c ↓ # of classes

n ↳ # of attributes

r → domain size of each attribute

$$\mathcal{LL}(\theta) = \log \left[\prod_{i=1}^{m} P(y^{(i)}; \Phi)\, P(x^{(i)} \mid y^{(i)}; \theta) \right]$$

$$= \sum_{i=1}^{m} \log P(y^{(i)}; \Phi) + \sum_{i=1}^{m} \sum_{j=1}^{n} \log P(x_j^{(i)} \mid y^{(i)}; \theta)$$

$x_j \perp x_k \mid y \Rightarrow P(x_1 \,--\, x_n \mid y)$

$\forall j,k$

$= \prod_j P(x_j \mid y)$

$$= \sum_{i=1}^{m} \sum_{c=1}^{C} \underbrace{1\{y^{(i)}=c\}}_{\text{indicator fn.}} \log \phi_c + \sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{c=1}^{C} 1\{y^{(i)}=c\} \log [p(x_j^{(i)} | y^{(i)}=c; \theta_{j|y=c})]$$

$$= \sum_{i=1}^{m} \sum_{c=1}^{C} \underbrace{1\{y^{(i)}=c\} \log \phi_c}_{T1} + \sum_{i=1}^{m} \sum_{j=1}^{n} \sum_{c=1}^{C} \underbrace{1\{y^{(i)}=c\} \sum_{\ell=1}^{r} 1\{x_j^{(i)}=\ell\} \log \theta_{j\ell|c}}_{T2}$$

$$\nabla_\Theta LL(\Theta) = 0$$

$$\Leftrightarrow \quad \underbrace{\nabla_\Phi LL(\Theta) = 0}_{T1} \qquad \underbrace{\nabla_\theta LL(\Theta) = 0}_{T2}$$

$$\nabla_{\phi_c} LL(\Theta) = 0 \qquad \forall c \in \{1 \dots C\}$$

↳ only T1 contributes to this gradient

$$\frac{\partial}{\partial \phi_c} LL(\Theta) =$$

$C-1$

$$\hookrightarrow \quad \sum_{i=1}^{m}\left[\frac{\partial}{\partial \phi_c} \log \phi_c\right] 1\{y^{(i)}=c\} + \sum_{i=1}^{m}\frac{\partial}{\partial \phi_c}\left[\log\Big(1-\sum_{c'=1}\phi_{c'}\Big)\right] 1\{y^{(i)}=c\}$$

$$= \sum_{i=1}^{m} \frac{1}{\phi_c} + \frac{1}{\left[1-\sum_{c'}\phi_{c'}\right]}(-1)$$

$$= \sum_{i=1}^{m} \frac{1\{y^{(i)}=c\}}{\phi_c} + (-1)\sum_{i=1}^{m} \frac{1\{y^{(i)}=0\}}{\underbrace{\left[1-\sum_{c'}\phi_{c'}\right]}_{\to\ \phi_0}}$$

Equate to zero

$$\Rightarrow \sum_{i=1}^{m} \frac{1\{y^{(i)}=c\}}{\phi_c} = \sum_{i=1}^{m} \frac{1\{y^{(i)}=0\}}{\phi_0}$$

$$\Rightarrow \frac{\phi_c}{\phi_{c_0}} = \frac{\sum_{i=1}^{m} 1\{y^{(i)}=c\}}{\sum_{i=1}^{m} 1\{y^{(i)}=c_0\}} \quad \forall c \quad — (1)$$

$$\Rightarrow \frac{\phi_c}{\phi_{c'}} = \frac{\sum_{i=1}^{m} 1\{y^{(i)}=c\}}{\sum_{i=1}^{m} 1\{y^{(i)}=c'\}} \quad — (2)$$

$$\Rightarrow \phi_c \sum_{i=1}^{m} 1\{y^{(i)}=c'\} = \phi_{c'} \sum_{i=1}^{m} 1\{y^{(i)}=c\} \quad c \neq c'$$

sum over all c' $\Rightarrow$

$$\phi_c \cdot m = \sum_{i=1}^{m} 1\{y^{(i)}=c\}$$

$$\phi_c = \frac{1}{m} \sum_{i=1}^{m} 1\{y^{(i)} = c\} \quad \text{—(A)}$$

In similar manner

$$\forall j, l, c \quad \theta_{jl|c} = \frac{\sum_{i=1}^{m} 1\{y^{(i)} = c\}\, 1\{x_j^{(i)} = l\}}{\sum_{i=1}^{m} 1\{y^{(i)} = c\}} \quad \text{—(B)}$$

MLE (Maximum likelihood estimate)